

SHARVARI CHUNNE

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PROFESSIONAL SUMMARY

Mechanical Engineer with expertise in CAD modeling (SolidWorks, CSWA Certified), additive manufacturing (DfAM), FEA/FEM structural validation, and design-for-manufacturing (DFM). Demonstrated success achieving 40% structural weight reduction and 20% performance improvement through topology optimization. Experienced in thermodynamic system modeling and verification and validation (V&V) using MATLAB. Strong foundation in Lean Manufacturing and Six Sigma methodologies. Currently completing M.S. in Mechanical Engineering (GPA: 3.69/4.0) with hands-on teaching and research experience supporting 50+ students.

CORE COMPETENCIES

CAD Modeling (SolidWorks, CATIA, AutoCAD, Fusion 360) | Additive Manufacturing & 3D Printing | FEA/FEM Analysis (ANSYS) | GD&T | Design for Manufacturing (DFM/DfAM) | Topology Optimization | MATLAB Simulation | Thermodynamic Modeling | Verification & Validation (V&V) | Lean Manufacturing | Six Sigma | Root Cause Analysis | BOM | CAM Machining | CNC Machining | Machine Tools | Technical Report Writing

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant | Cleveland State University, Cleveland, OH January 2025 – May 2026

- Delivered instructional support for ESC 120 (Introduction to Engineering Design), MCE 515/493 (Additive Manufacturing), MCE 481 (Thermodynamics Lab), and ESC 282 (Engineering Economy).
- Mentored 50+ students in SolidWorks CAD modeling, design calculations, laboratory experimentation, data analysis, and technical report writing.
- Evaluated student design projects using structured grading rubrics and provided actionable feedback to improve engineering comprehension.
- Supported laboratory setup, equipment troubleshooting, and safety compliance for hands-on engineering coursework.

PROJECTS

Heat Pump Waste Heat Recovery System – Modeling and V&V October 2025 – December 2025

- Developed and validated a MATLAB-based thermodynamic model using real operational data to assess system performance and energy efficiency.
- Performed verification, validation (V&V), and parametric analysis; documented assumptions, uncertainty analysis, and engineering conclusions following standard modeling and simulation practices.

Comparative Study of Additive Manufacturing in Automotive Components January 2025 – May 2025

- Bugatti Brake Caliper vs. GM Seat Bracket: applied DfAM principles and topology optimization.
- Achieved **40% structural weight reduction** and **20% performance improvement** through material selection and design optimization.
- Conducted FEA-based structural validation using ANSYS; compared additive manufacturing vs. conventional manufacturing methods for cost, performance, and material efficiency.

Lakewood Sustainable Energy Community Development August 2024 – December 2024

- Designed a renewable energy integration strategy achieving **59% reliance on sustainable energy sources** for municipal applications.
- Performed feasibility analysis of solar, wind, and hybrid energy solutions; evaluated lifecycle cost, performance metrics, and environmental impact.

Assembly of HOLMATRO 4240 Spreader Rescue Tool

April 2023 – May 2023

- Assembled hydraulic and mechanical components following technical drawings and safety standards.
- Analyzed pressure mechanisms and load-bearing components to verify functional performance and operational reliability.

Solar Panel Selection Using Multi-Criteria Decision-Making (MCDM)

August 2021 – June 2022

- Applied AHP (Analytic Hierarchy Process) and TOPSIS methodologies to evaluate solar panel alternatives based on cost, durability, and performance criteria.
- Developed weighted decision matrices and conducted quantitative trade-off analysis to optimize system selection accuracy.

TECHNICAL SKILLS

CAD Software: SolidWorks (CSWA Certified), CATIA, AutoCAD, Fusion 360, GrabCAD

Simulation & Analysis: ANSYS (FEA/FEM), MATLAB, Thermodynamic Modeling, Parametric Analysis

Manufacturing: 3D Printing, Additive Manufacturing, CNC Machining, CAM, DfAM, Post-Processing, Slicing Software

Design & Engineering: Mechanical Design, GD&T, DFM, FEA/FEM, BOM, Machine Tools, Design Calculations

Process & Quality: Lean Manufacturing, Six Sigma, Root Cause Analysis, Continuous Improvement

Productivity: Microsoft Office (Word, Excel, PowerPoint)

EDUCATION

M.S. in Mechanical Engineering | Cleveland State University, Cleveland, OH

August 2024 – May 2026

GPA: 3.69 / 4.0

B.Tech in Mechanical Engineering | Government College of Engineering, India

August 2018 – July 2022

Distinction

CERTIFICATIONS

Modeling and Simulation in Engineering (Fundamentals)

June 2026

Machine Tool – CNC Machining (Fusion 360), IACMI – The Composites Institute

May 2026

Six Sigma and Lean Manufacturing

July 2025

Machine Design on CAD Application in Design Thinking

July 2023

Certified SolidWorks Associate (CSWA) – Mechanical Design

June 2023

SolidWorks PDM Workshop

June 2023